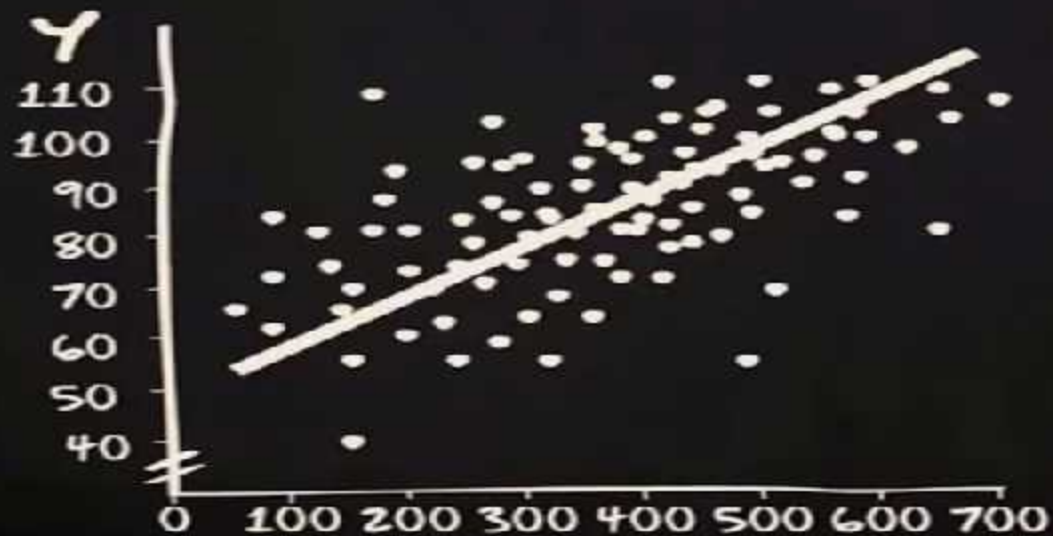


PEARSON'S R

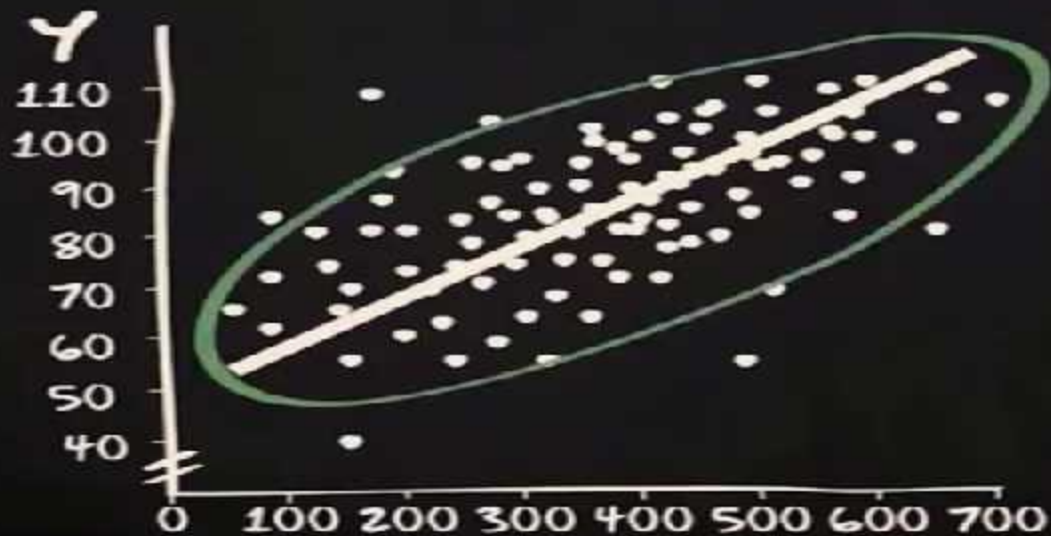
direction and strength of linear correlation with one number





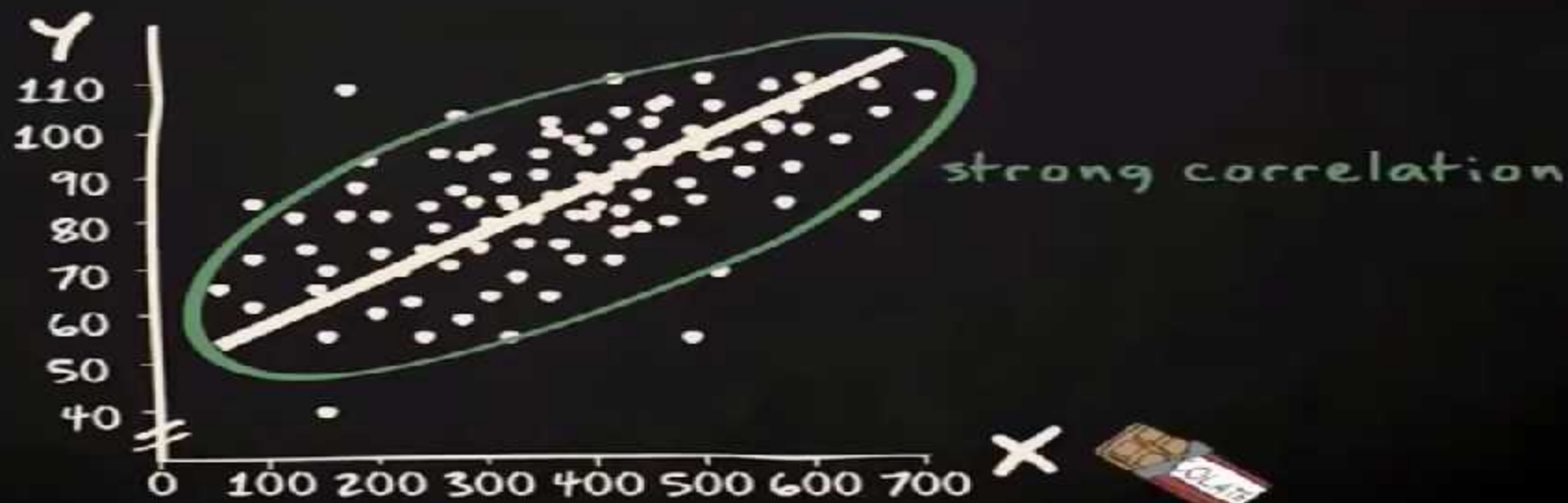
X

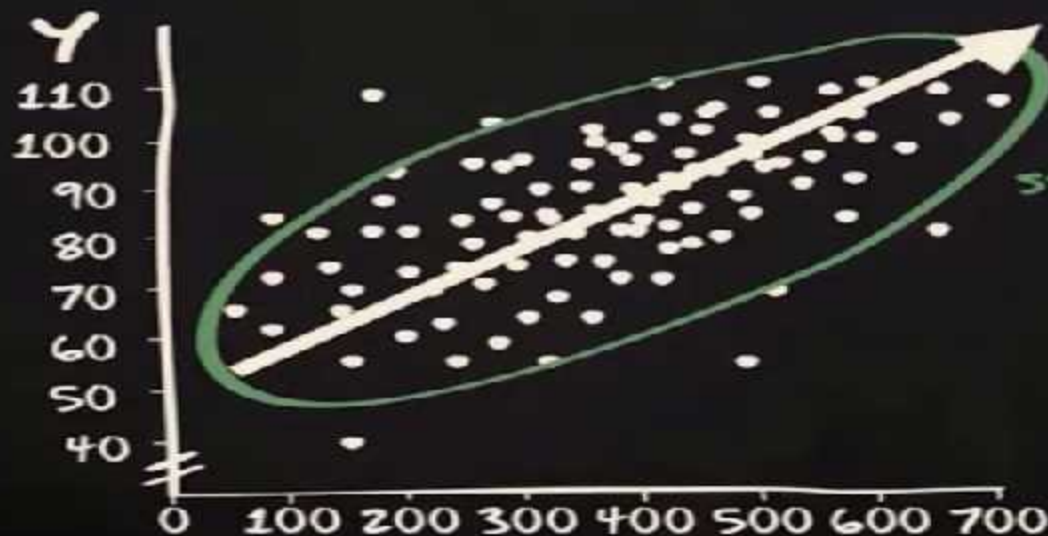




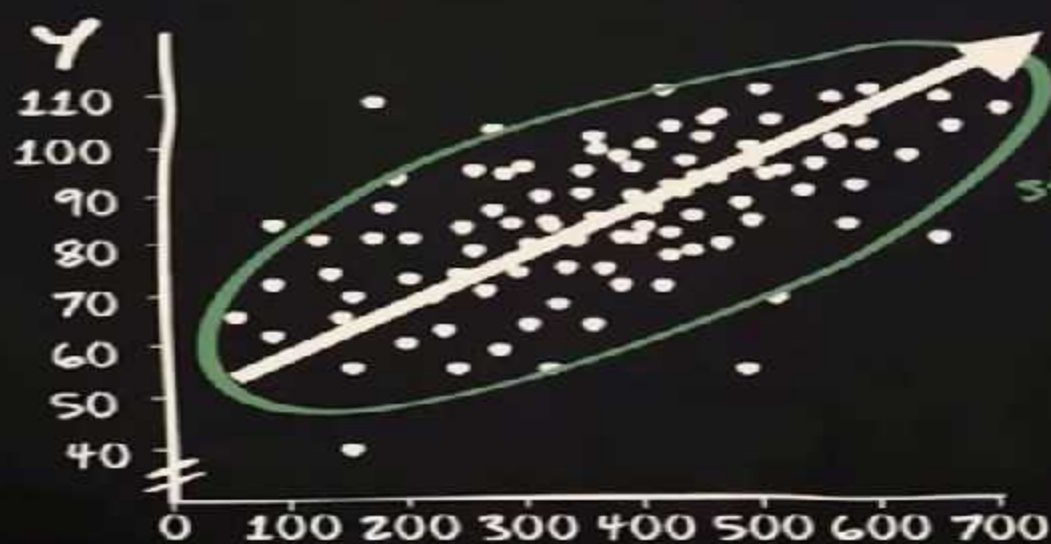
X







strong correlation



positive correlation

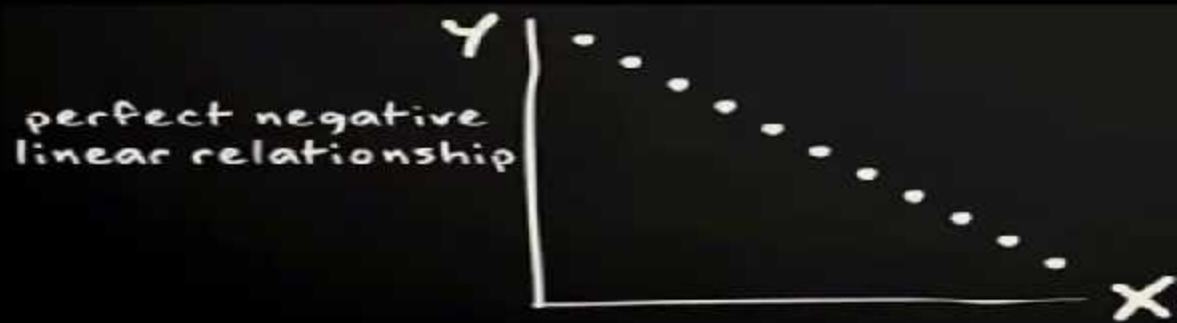
strong correlation

X





CONCLUSION
strong positive
linear
relationship



SCATTERPLOT



strong or weak
correlation?

PEARSON'S R



HOW
strong or weak
correlation?

SCATTERPLOT



strong or weak
correlation?

PEARSON'S R



HOW
strong or weak
correlation?

direction

+ = positive

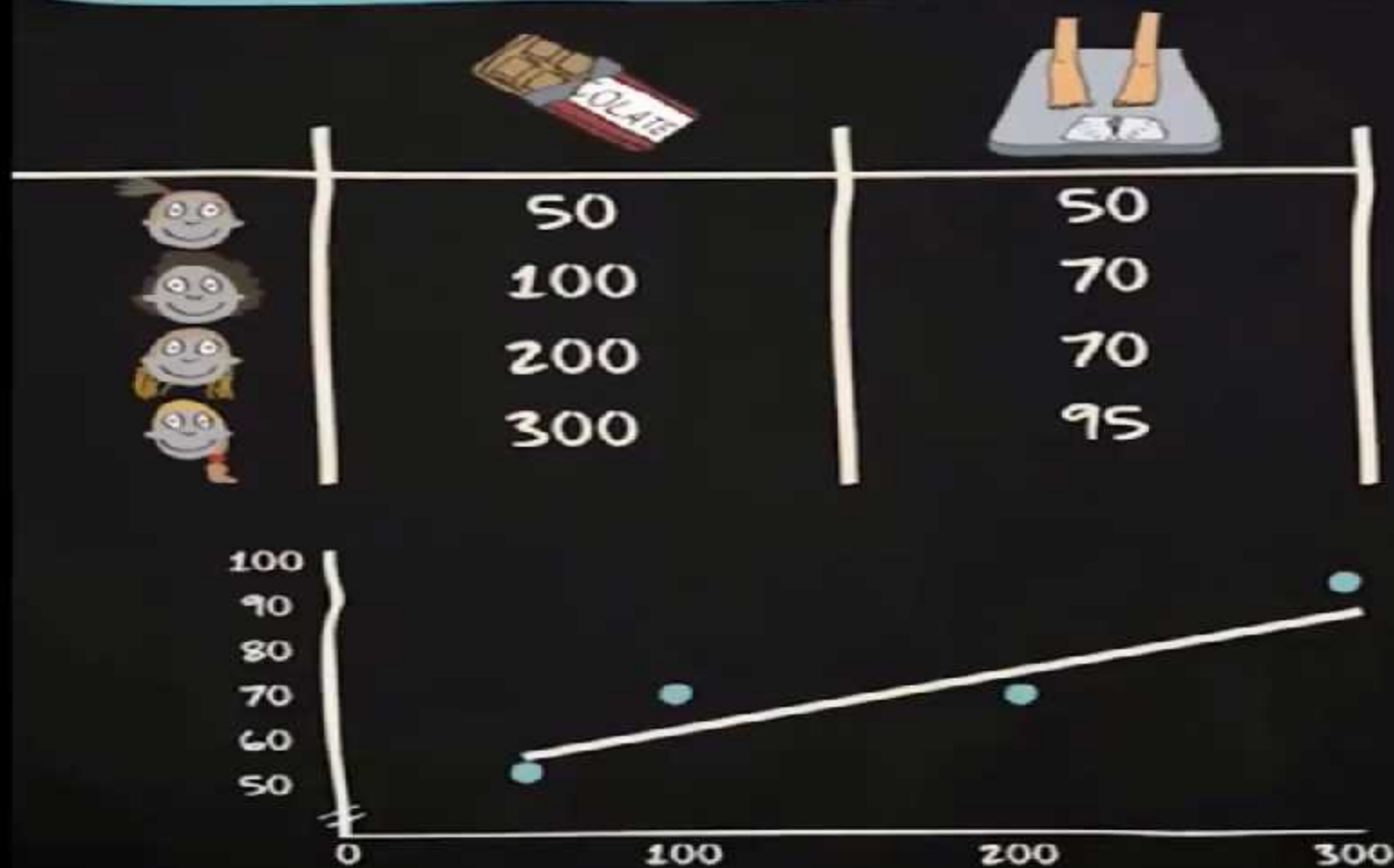
- = negative

strength

-1 = perfect negative

+1 = perfect positive

COMPUTE PEARSON'S R



COMPUTE PEARSON'S R



50
100
200
300

50
70
70
95



COMPUTE PEARSON'S R



50	50
100	70
200	70
300	95

100
90
80
70
60
50

0

100

200

300

COMPUTE PEARSON'S R

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

COMPUTE PEARSON'S R

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

Mean



x



y

COMPUTE PEARSON'S R

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

	 X	 Y
Mean	162.5	71.3
SD	110.9	18.4

COMPUTE PEARSON'S R

$$z = \frac{x - \bar{x}}{s} = \frac{50 - 162.5}{110.9}$$

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

Mean
SD



X

162.5
110.9



Y

71.3
18.4

X

50
100
200
300

Z_x

Y

50
70
70
95

Z_y

Z_x * Z_y



COMPUTE PEARSON'S R

$$z = \frac{x - \bar{x}}{s} = \frac{50 - 162.5}{110.9} = -1.01$$

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

Mean
SD

X
162.5
110.9

Y
71.3
18.4

	X 	Z _x	Y 	Z _y	Z _x *Z _y
	50	-1.01	50		
	100	-0.56	70		
	200	0.34	70		
	300	1.24	95		

COMPUTE PEARSON'S R

$$z = \frac{x - \bar{x}}{s} = \frac{50 - 162.5}{110.9} = -1.01$$

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

Mean
SD

X
162.5
110.9

Y
71.3
18.4

	X	Z _x	Y	Z _y	Z _x *Z _y
☹️	50	-1.01	50	-1.15	
☹️	100	-0.56	70	-0.07	
☹️	200	0.34	70	-0.07	
☹️	300	1.24	95	1.29	

COMPUTE PEARSON'S R

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

Mean

SD

X

162.5

110.9

Y

71.3

18.4

X

50

100

200

300

Z_x

-1.01

-0.56

0.34

1.24

Y

50

70

70

95

Z_y

-1.15

-0.07

-0.07

1.29

Z_x * Z_y

COMPUTE PEARSON'S R

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

Mean

SD

X

162.5

110.9

Y

71.3

18.4

X

50

100

200

300

Z_x

-1.01

-0.7

0.7

1.24

Y

50

70

70

95

Z_y

-1.15

-0.7

-0.7

1.1

Z_x*Z_y

-1.01 * -1.15

COMPUTE PEARSON'S R

$$r = \frac{\sum Z_x Z_y}{n - 1}$$

Mean

SD

X

162.5

110.9

Y

71.3

18.4

X

50

100

200

300

Z_x

-1.01

-0.56

0.34

1.24

Y

50

70

70

95

Z_y

-1.15

-0.07

-0.07

1.29

Z_x*Z_y

1.17

0.04

-0.02

1.60

COMPUTE PEARSON'S R

$$r = \frac{\sum Z_x Z_y}{n-1}$$

Mean
SD

X
162.5
110.9

Y
71.3
18.4

	X	Z _x	Y	Z _y	Z _x *Z _y
😊	50	-1.01	50	-1.15	1.17
😊	100	-0.56	75	-0.07	0.04
😊	200	0.34	70	-0.07	-0.02
😊	300	1.24	95	1.29	1.60
					<u>2.78</u>

COMPUTE PEARSON'S R

$$r = \frac{2.78}{4-1}$$

Mean

SD

X

162.5

110.9

Y

71.3

18.4

X

50

100

200

300

Z_X

-1.01

-0.56

0.34

1.24

Y

50

70

70

95

Z_Y

-1.15

-0.07

-0.07

1.29

Z_X*Z_Y

1.17

0.04

-0.02

1.60

2.78

COMPUTE PEARSON'S R

$$r = \frac{2.78}{4-1}$$

$$r = 0.93$$

COMPUTE PEARSON'S R

$$r = \frac{2.78}{4-1}$$

$$r = 0.93$$

What does this mean?

COMPUTE PEARSON'S R

$$r = \frac{2.78}{4-1}$$

$$r = 0.93$$

What does this mean?



Strong positive
linear relationship
between



&



important note



check scatterplot
before you calculate
Pearson's r

important note



check scatterplot
before you calculate
Pearson's r

NO
linear relation



NO
Pearson's r

important note



check scatterplot
before you calculate
Pearson's r

NO
linear relation



NO
Pearson's r



important note



check scatterplot
before you calculate
Pearson's r

NO
linear relation



NO
Pearson's r

